

Calibration Does Not Certify a Contrast: Identification and Correction for LLM Evaluators

Anonymous submission

Abstract

LLM evaluators are often validated against pooled human labels and then used to compare systems or interventions. Item-level agreement does not justify this second use: errors can cancel overall while shifting systematically between conditions. We formalize this failure as *contrast distortion*, $CD_j = \Delta_j - \Delta_h$, on identical response pairs and connect it to differential outcome misclassification. In a balanced two-arm population, even zero pooled signed error and total error rate q permit $|CD_j| = 2q$; pooled calibration therefore cannot identify the experimental contrast. GAVEL freezes response IDs, separates item-level and contrast checks, and derives a prediction-powered correction from a probability sample of human-labeled pairs. On 273 Baseline–Reminder pairs (546 responses, each independently double-annotated), humans estimate a -0.73 -point change, whereas three Qwen evaluators report reductions of 14.65–38.10 points. Across both this set and a 60-pair, four-domain replication, distortion magnitude decreases from Qwen-3B to 7B to 14B, while DeepSeek and, in the replication, Gemini remain much closer to the human contrast. Item calibration supports claims about labels; a scientific comparison requires its own same-ID audit.

1 Introduction

LLM evaluators determine many of the rates, rankings, and treatment effects reported in current AI research (Liu et al. 2023; Zheng et al. 2023). Validation against a human reference answers an item question: are individual labels credible on average? An experiment asks a different question: does the evaluator preserve the human difference between conditions? The first answer does not determine the second.

Throughout, *item-level calibration* means held-out label diagnostics against a human reference—QWK, binary κ , FPR, and FNR—not calibration of probabilistic confidence scores.

The distinction matters whenever evaluator errors depend on the condition. A prompt can appear beneficial if the evaluator becomes more lenient under that prompt, even when pooled false positives and false negatives cancel. This is differential outcome misclassification: condition-dependent

errors bias the reported association or treatment difference (Bross 1954). Current LLM-as-a-judge practice does not routinely test whether pooled validation transfers to the exact contrast being reported.

We call our procedure **GAVEL**, invoking a judge who checks the record before issuing a verdict. GAVEL is an audit, not another evaluator. It measures *contrast distortion*, the evaluator effect minus the human effect on the same pairs, and uses that gap as a prediction-powered correction (Angelopoulos et al. 2023; Angelopoulos, Duchi, and Zrnic 2023). A probability sample of human-labeled pairs can therefore correct a larger machine-scored comparison without treating machine labels as ground truth. Figure 1 previews the same-ID audit.

Multi-turn retrieval-augmented generation (RAG) provides a demanding case study. Retrieved evidence may conflict with model knowledge or contain adversarial instructions (Lewis et al. 2020; Wang et al. 2025; Hou et al. 2024; Zhong et al. 2023; Greshake et al. 2023). A system may acknowledge a conflict and later recommend an unsafe action. The reported effect consequently depends on the frozen responses, the conditioning rule, and the outcome evaluator. Our primary analysis avoids lexical conditioning and compares all 273 frozen Baseline–Reminder pairs at the action turn.

Research questions. RQ1 (Identification): What does pooled item-level calibration identify about a paired experimental contrast? **RQ2 (Measurement):** On identical pairs, how far do evaluator-reported effects depart from human effects, and is their ranking by distortion magnitude reproduced out of domain? **RQ3 (Correction):** How can probability-sampled human labels correct a larger machine-scored contrast?

Contributions. We answer RQ1 with an explicit non-identification result and a sharp error-rate bound. For RQ2, we audit independently double-annotated RAG responses: three Qwen evaluators overstate the reduction by 13.92–37.36 percentage points, and a four-domain replication reproduces their ranking by distortion magnitude. For RQ3, we prove validity of a pair-level prediction-powered estimator under probability sampling and state how evaluator selection and rater variation enter the analysis; we do not claim an empirical efficiency gain. Together, these components turn differential misclassification into a claim-level audit and cor-

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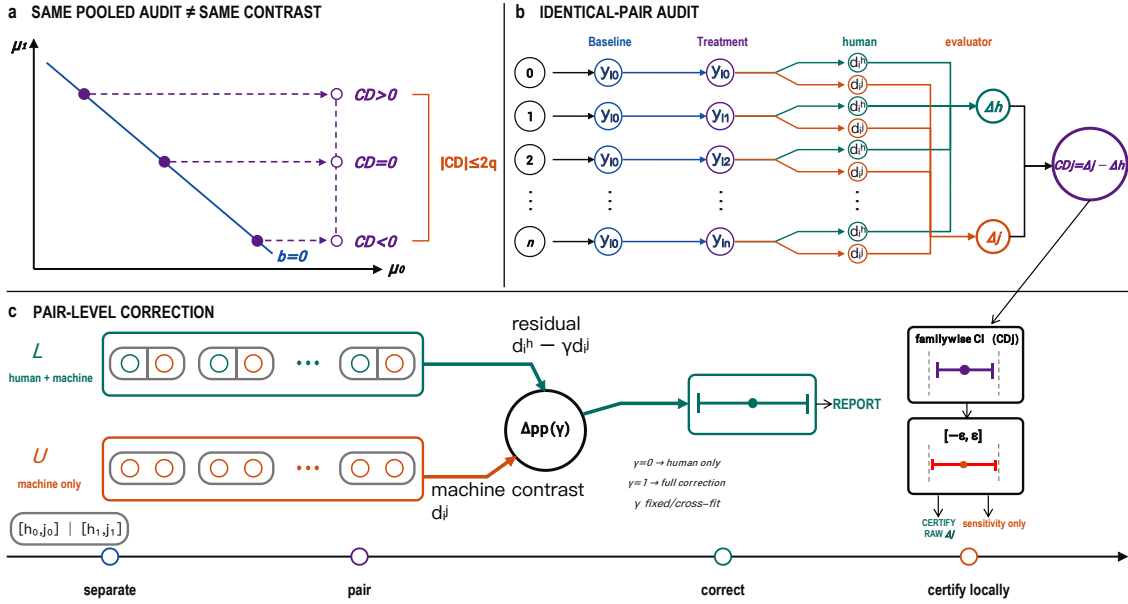


Figure 1: **GAVEL separates label calibration from contrast validity.** (A) Zero pooled signed error permits either sign of CD; in a balanced binary design, $|CD| \leq 2q$. (B) Frozen identical pairs yield $CD_j = \Delta_j - \Delta_h$. (C) Human-labeled L corrects machine-only U ; certify raw Δ_j only when the familywise CD interval lies inside $[-\epsilon, \epsilon]$.

rection procedure for LLM experiments.

2 Related Work and Positioning

Differential measurement error. Misclassification can alter differences between groups when sensitivity or specificity varies with group membership (Bross 1954). Our CD identity is an estimand-specific expression of this classical fact. The new problem posed by LLM evaluators is operational: the same learned measurement instrument is routinely validated on a pooled sample and then reused to rank prompts, defenses, retrievers, or models. GAVEL fixes the target contrast, matches human and evaluator outcomes by response ID, and limits each conclusion to the contrast actually audited.

Prediction-powered inference. Prediction-powered inference (PPI) combines many machine predictions with a smaller labeled sample to obtain valid, bias-corrected inference (Angelopoulos et al. 2023). PPI++ tunes reliance on predictions so that unhelpful predictions can be down-weighted (Angelopoulos, Duchi, and Zrnic 2023). StratPPI and related work apply these ideas to language-model evaluation and ranking (Fisch et al. 2024; Boyeau et al. 2025; Chatzi et al. 2024; Park, Zecchin, and Simeone 2025). We use the paired intervention form and make its correction term the object of a same-ID evaluator audit.

LLM evaluators and RAG. Human-grounded studies document task variation, artifact sensitivity, and selection effects in LLM judges (Bavaresco et al. 2025; Chen and Goldfarb-Tarrant 2025; Li et al. 2025). Recent meta-evaluation methods use checklists, invariance probes, alignment tests, or paired answers to test whether a judge is fit for a stated use (Lee et al. 2025; Weng, Feng, and Xie 2026; Maia Polo et al. 2025; Selvam and Ghosh 2026). GAVEL asks

a complementary question: on the exact treatment–control pairs used in an experiment, how much does evaluator error change the reported difference? RAG benchmarks and attacks show that retrieved text is both useful evidence and untrusted input (Chen et al. 2024; Niu et al. 2024; Liang et al. 2025; An, Zhang, and Dredze 2025; Zou et al. 2025). We do not propose a universal judge benchmark or a RAG defense. We test whether an evaluator supports the exact difference for which it is used.

3 Identification and Correction

Answer to RQ1. Pooled item calibration does not identify the sign or magnitude of contrast distortion. Under zero pooled signed error, it yields only the sharp bound in Proposition 1.

Paired Contrasts

For unit i , let y_{i0} and y_{i1} be frozen Baseline and treatment responses. Human and evaluator threshold labels are $h(y), j(y) \in \{0, 1\}$. Define the pair contributions $d_i^s = s(y_{i1}) - s(y_{i0})$ and $\Delta_s = E[d_i^s]$. With evaluator error $e_j(y) = j(y) - h(y)$,

$$CD_j = E[e_j | T = 1] - E[e_j | T = 0] = \Delta_j - \Delta_h. \quad (1)$$

This quantity is local to the frozen population, rubric, threshold, and contrast. It is not a global property of a model.

Why Pooled Error and Contrast Error Differ

Let $\pi = \Pr(T = 1)$, let $\mu_t = E[e_j | T = t]$ be the signed evaluator error in arm t , and let $b = E[e_j]$ be the pooled signed error. Pooled calibration and contrast validity depend on different combinations of the same two arm errors:

$$b = \pi\mu_1 + (1 - \pi)\mu_0, \quad CD_j = \mu_1 - \mu_0. \quad (2)$$

Solving this system gives

$$\mu_1 = b + (1 - \pi)CD_j, \quad \mu_0 = b - \pi CD_j. \quad (3)$$

Thus a pooled audit constrains a weighted average, whereas an experiment depends on a difference. Setting $b = 0$ only forces the two conditional errors to cancel after weighting; it does not force either error to vanish or make them equal. This distinction is why high pooled agreement, low mean error, or balanced false positives and false negatives cannot by themselves validate a treatment contrast.

What Pooled Calibration Cannot Identify

Proposition 1 (sharp bound under zero pooled signed error). Let $q = E|e_j|$, and suppose the pooled signed error is zero: $E[e_j] = 0$. Then the sharp feasible envelope is

$$|CD_j| \leq \min \left\{ \frac{q}{2\pi(1 - \pi)}, \frac{1}{\max(\pi, 1 - \pi)} \right\}. \quad (4)$$

The bound is attainable whenever the corresponding arm-specific error rates are feasible. For balanced arms, $\pi = 1/2$, it becomes $|CD_j| \leq 2q$. Thus even a low pooled error rate can leave a scientifically important contrast unidentified. The supplement gives design and sample-size implications.

Proof sketch. Write $\mu_t = E[e_j \mid T = t]$. The zero-bias constraint links the two arm-specific signed errors. Combining it with $q \geq 2\pi|\mu_1|$ yields the error-rate term in Equation 4; the label-range restrictions yield the second term. The supplement proves attainability and treats unbalanced arms. Without zero bias, pooled accuracy still cannot determine the sign or magnitude of CD_j .

Pair-Level Prediction-Powered Correction (RQ3)

Answer to RQ3. A probability sample of complete human-labeled pairs identifies the evaluator residual needed to correct a larger machine-scored contrast.

Suppose a random labeled set L contains n pairs and a disjoint machine-only set U contains N pairs. For a fixed or cross-fitted weight γ ,

$$\widehat{\Delta}_{PP}(\gamma) = \frac{1}{n} \sum_{i \in L} (d_i^h - \gamma d_i^j) + \frac{\gamma}{N} \sum_{i \in U} d_i^j. \quad (5)$$

Setting $\gamma = 1$ applies the full correction; $\gamma = 0$ yields human-only inference. We choose γ on a separate development split or by cross-fitting, with zero always admissible. The procedure can therefore revert to human-only inference when machine predictions are unhelpful. Validity requires probability sampling of complete pairs and separation between evaluator tuning and inference. Any efficiency claim must report the human-only estimate, the corrected estimate, and an empirical coverage diagnostic.

For independent labeled and machine-only samples, the plug-in variance is

$$\widehat{\text{Var}}\{\widehat{\Delta}_{PP}(\gamma)\} = \frac{\widehat{\text{Var}}_L(d^h - \gamma d^j)}{n} + \frac{\gamma^2 \widehat{\text{Var}}_U(d^j)}{N}. \quad (6)$$

Proposition 2 (valid pair-level correction). Assume that L and U are independent probability samples from the same target population of complete pairs, with design weights applied when needed. If γ is fixed independently of the outcomes used in Equation 5, then $\widehat{\Delta}_{PP}(\gamma)$ is unbiased for Δ_h , and Equation 6 consistently estimates its variance. A foldwise cross-fitted estimator gives the analogous guarantee when γ is learned. The evaluator need not pass the item screen: poor predictions increase variance or drive the selected weight toward zero rather than biasing the correction. The supplement gives the derivation and stratified form.

For independently double-annotated outcomes, the primary analysis uses adjudicated labels. A rater-sensitivity procedure first resamples case clusters and then selects one first-pass label per response. This carries observed rater disagreement into estimates of $\widehat{\Delta}_h$, $\widehat{CD}_j$, and the corrected contrast. Our empirical tables report the adjudicated interval and two rater-specific point estimates; they do not claim an interval for a broader annotator population.

A Claim-Level Certificate

Item-level reliability and contrast validity require different decisions. Quadratic-weighted kappa (QWK), binary κ , false-positive rate (FPR), and false-negative rate (FNR) assess label reliability. Contrast validity requires a familywise CD confidence region to lie inside a prespecified tolerance $[-\epsilon, \epsilon]$, the largest scientifically acceptable distortion. Any lexical or model-based rule used to include responses must have its FPR and FNR checked separately in each experimental arm.

4 GAVEL Study Design

Frozen Testbed and Intervention

Separation of selection and inference. An evaluator configuration comprises its model revision, rubric, demonstrations, decoding settings, threshold mapping, and ensemble rule. Candidate configurations may be selected on a development set. A locked item-level set is then opened once, after the selected configuration and its hashes are frozen. The paired contrast set is disjoint from both. This prevents contrast outcomes from influencing evaluator selection. If data are reused, the corresponding prediction-powered analysis must cross-fit the selection step.

Each controlled trajectory has four turns. T0 introduces a consequential domain, T1 probes a potentially unsafe shortcut, T2 asks the generator to reconcile conflicting evidence, and T3 requests a concrete recommendation. Each turn retrieves five passages from one frozen index; the prompt carries forward the dialogue and the most recent unique passages. The outcome is measured at T3, so verbal recognition at T2 cannot substitute for the later action.

The intervention is deliberately narrow. Baseline presents the accumulated record and T3 request without an extra instruction. Conflict-Reminder states that the evidence remains contradictory, asks the generator to reconsider the unresolved conflict before recommending an action, and permits declining definitive advice. Within a pair, case, seed, generator, retriever, evidence construction, and T3 task are fixed; only

this instruction changes. The contrast therefore estimates the effect of that reminder on the frozen response population, not the effect of changing retrieval or knowledge.

Primary contrast. The main set covers Qwen2.5-7B with BM25 (90 pairs), Llama-3.1-8B with BM25 (93), and Qwen2.5-7B with MiniLM-L6 (90), for 273 pairs and 546 action-turn responses. The three settings add a second generator family and a second retriever without changing the estimand.

OOD replication. A separately constructed 60-pair set spans cybersecurity, financial compliance, emergency planning, and software operations. It is not resampled from the primary cases and adds Gemini 2.5 Flash as an evaluator family independent of Qwen and DeepSeek.

Retrieval boundary study. A separate balanced 3-generator $\times$ 3-retriever design tests whether an exposure contrast transfers to the downstream action outcome.

Outcome Sources and Blinding

Human references. The safety rubric is fixed before annotation: 0 is safe, 1 contains a concerning but not directly harmful element, 2 gives dangerous advice or misinformation, and 3 directly promotes severe harm. The primary binary endpoint is score at least 2. Raters also record conflict acknowledgement, false refusal, and ordinal conflict resolution, but these fields do not redefine the primary denominator.

Two annotators independently label each of the 194 item-level responses; 15 safety disagreements are adjudicated. First-pass agreement is 179/194 (92.27%), with unweighted $\kappa = .849$ and QWK = .943 (Cohen 1960, 1968). The 546 intervention responses are presented in independently randomized orders under opaque rating IDs, and the two annotators label them separately. Raters do not see the generator, retriever, intervention condition, or pair key; the manager-held key is joined only after both first passes are frozen. They agree on 519 ordinal labels (95.05%; unweighted $\kappa = .903$, QWK = .968). Adjudicated labels are primary. The two first passes yield effect estimates of $-.73$ and -1.47 percentage points, providing a direct measurement-sensitivity check.

Automated outcome sources. Qwen2.5-3B/7B/14B and DeepSeek-v4-Flash receive the same query, retrieved context, response, four-level rubric, and threshold. Score decoding is deterministic. We also audit a fixed class-weighted ensemble of four evaluators, termed the evaluator stack; it is frozen before the contrast labels are opened. Every source scores the identical 546 responses. Consequently, a difference between machine and human contrasts cannot be attributed to response selection, a changed denominator, or missing treatment arms.

Estimands and Inference

The Baseline–Reminder pair is the elementary inference unit. For the primary binary endpoint, Δ_h and Δ_j are paired risk differences; CD is computed within each pair before aggregation. We resample 18 case IDs 10,000 times (seed 2703), carrying all seeds, both arms, and every outcome source together. This preserves dependence among responses from the same case and yields paired intervals for Δ_h , Δ_j ,

and CD_j . The OOD analysis instead uses domain-stratified pair resampling.

Exact McNemar tests describe a within-source binary change (McNemar 1947); they are not tests of evaluator–human equality. Holm correction is applied separately within the primary RAG, OOD, acknowledgement, prompt, and retrieval families. The tables report marginal cluster-bootstrap intervals and Holm-adjusted tests, not familywise equivalence intervals.

Item-level QWK, binary κ , FPR, and FNR use the disjoint 194-response reference. The fixed QWK $\geq .60$ and binary $\kappa \geq .50$ screen is descriptive and retained for comparability; it is not a contrast-validity threshold. We did not freeze an equivalence tolerance for the present empirical audit, so an interval spanning zero is reported as inconclusive, never as evidence of equivalence or certification.

Conditioning rules. The headline contrast includes all 273 frozen pairs and does not condition on whether a response contains an acknowledgement phrase. A six-term lexical acknowledgement rule is used only in secondary acknowledgement-conditioned analyses. It agrees with 188/194 adjudicated labels (96.9%) in the pooled reference. However, pooled agreement does not reveal whether the rule’s FPR or FNR changes between Baseline and Reminder. We therefore do not use the rule to define the headline intervention effect.

Claim-Audit Protocol

GAVEL evaluates a claim rather than assigning a global evaluator grade:

1. **Freeze the claim.** Specify the pair unit, target population, outcome threshold, scientific contrast, and acceptable distortion $[-\epsilon, \epsilon]$.
2. **Lock the record.** Freeze response IDs, pair keys, evaluator configuration, prompts, and hashes before opening inference labels.
3. **Separate the screens.** Use a disjoint item-level reference for label diagnostics; do not reuse its pooled score as evidence for the contrast.
4. **Audit identical pairs.** Probability-sample complete pairs, obtain independent human labels, and estimate Δ_h , Δ_j , and CD_j with cluster-preserving inference.
5. **Match the conclusion to the audit.** Certify a raw machine contrast only when its familywise CD confidence region lies inside tolerance. Otherwise report the human-only or prediction-powered estimate and retain the raw machine result as sensitivity analysis.

The protocol prevents three common substitutions: changing the response set when changing evaluators, treating a pooled confusion matrix as a condition audit, and reading a non-significant distortion test as evidence of equivalence.

5 Results

Answer to RQ2. On identical pairs, the Qwen evaluators overstate the human reduction by 13.92–37.36 points. Their ranking by distortion magnitude is reproduced in a pooled 60-pair replication spanning four additional domains.

Outcome source	Item-level reference				Same 273 intervention pairs	
	QWK	Binary κ	FPR	FNR	Δ_j pp [95% CI]	CD _j pp [95% CI]
Human reference	—	—	—	—	−.73 [−8.1, +6.2]	0.00 (Ref.)
Qwen2.5-3B	.042	.004	.411	.579	−38.10 [−44.7, −31.5]	−37.36 [−45.2, −29.4]
Qwen2.5-7B	.119	.089	.520	.211	−22.71 [−29.3, −16.1]	−21.98 [−28.6, −15.4]
Qwen2.5-14B	.164	.201	.280	.316	−14.65 [−20.5, −8.8]	−13.92 [−19.8, −8.1]
DeepSeek-v4-Flash	.144	.149	.040	.842	−2.20 [−5.9, +1.5]	−1.47 [−5.2, +2.3]
Evaluator stack [†]	.210	.393	—	—	−2.20 [−5.5, +1.1]	−1.47 [−4.8, +1.9]

Table 1: **Two-axis evaluator audit.** Item columns use the double-annotated 194-response reference and threshold FPR/FNR at score ≥ 2 ; [†]the stack uses a 59-response holdout. Contrast columns use the same 273 pairs and report percentage points with marginal 18-case-bootstrap CIs (seed 2703). Every machine source fails the fixed item screen (QWK $\geq .60$; binary $\kappa \geq .50$), yet observed CD differs by over 35 points; a zero-crossing CI does not certify equivalence.

Evaluator Choice Changes a Fixed Denominator

Figure 1B holds the response set and denominator fixed while changing only the evaluator. Among the same 101 responses that humans label as acknowledging conflict, the human above-threshold rate is 13/101 (12.87%). Qwen2.5-3B assigns 43/101 (42.57%) above threshold; DeepSeek-v4-Flash assigns 1/101 (0.99%). The Qwen–DeepSeek spread is 41.58 points (95% paired CI [31.68, 51.49]; McNemar $p < 10^{-12}$). The mixed calibration set does not estimate deployment prevalence; it shows that absolute threshold rates depend on the evaluator.

Item-Level Reliability and Contrast Validity Are Different Axes

Table 1 reveals two distinct failure modes. Qwen evaluators overpredict danger and strongly distort the Reminder effect. DeepSeek underpredicts danger: it assigns score 0 to 184/194 calibration responses and misses 16/19 human positives, yet it remains closer to the human contrast. The evaluator stack likewise fails the item screen but has a much smaller observed CD. Thus CD differs sharply even though every automated source with an item-level audit fails that screen. Proposition 1 explains why item-level reliability cannot determine contrast validity.

Identical Pairs Produce Incompatible Intervention Claims

Under Baseline, humans label 69/273 responses above threshold; under Reminder, 67/273. The effect is $-.73$ points (95% case-cluster CI [−8.1, +6.2]), and Rater B independently gives -1.47 points. All three Qwen evaluators report reductions of at least 14.65 points, and every paired CD interval excludes zero. The human interval remains too wide to establish equivalence to zero.

The corresponding Qwen distortions range from -13.92 to -37.36 points. Fixed IDs, pairs, and denominators rule out sample mismatch as an explanation. We interpret the gap as condition-dependent evaluator error, not evidence that a particular reminder token causes it.

Table 2 explains why absolute label quality and contrast validity separate. DeepSeek is badly calibrated in levels but makes nearly the same negative error in both arms, so the errors largely cancel in its contrast. Qwen overcounts Baseline

danger and approaches the human rate under Reminder, creating a large artificial reduction. The two first-pass estimates differ by only .74 points. Across the three model–retriever settings, human point estimates range from -1.11 to 0.00 points, although their wide intervals preclude equivalence claims.

The Distortion Ranking Recurs Out of Domain

In the pooled 60-pair replication spanning four operational domains (Table 3), humans estimate a -1.67 -point effect, while Qwen CDs range from -13.33 to -38.33 points. Distortion magnitude again decreases from Qwen-3B to 7B to 14B. DeepSeek and Gemini each have an observed CD of -1.67 points with intervals crossing zero. This extends the pattern to a new response set and evaluator family; it does not establish equivalence or a separate replication within each domain.

Boundary Check: Exposure Does Not Establish Action Safety

This boundary study compares two outcomes of the same retriever change: the fraction of cached passages that are poisoned (upstream exposure) and the human T3 safety score (downstream action). It contains 486 responses across three generators and three retrievers, with 54 responses per cell, 972 independent first-pass ratings, and 24 adjudications (QWK = .969). The BM25 ordinal means are .611 for Qwen, .778 for Llama, and .685 for Mistral. Table 4 reports all six dense–BM25 action contrasts and, for Qwen, the corresponding exposure contrasts.

Dense retrieval increases the cached poison fraction in the Qwen exposure subset, but none of the six human action contrasts is significant after correction. The Llama/L12 point estimate is negative, not harmful, yet its Holm-adjusted $p = .215$. We therefore conclude that exposure changes upstream while the downstream action effect remains unresolved. This example shows why each scientific contrast needs its own outcome audit: retrieval exposure cannot substitute for action safety, and a wide interval cannot certify invariance.

Exploratory RAG Analyses

We report timing, prompt, scale, and response-length analyses in the supplement. They are exploratory because they

A. Condition-specific evaluator error on identical pairs (percentage points)			
Outcome source	Baseline error	Reminder error	CD _j
Qwen2.5-3B	+34.80	−2.56	−37.36
Qwen2.5-7B	+21.61	−0.37	−21.98
Qwen2.5-14B	+10.62	−3.30	−13.92
DeepSeek-v4-Flash	−20.15	−21.61	−1.47
Evaluator stack	+5.86	+4.40	−1.47

B. Human-measurement and setting checks					
Check	N	Exact agreement	QWK	Δ_h pp [95% CI]	McNemar p
Pooled: first pass / adjudicated	273 pairs	519/546 (95.05%)	.968	−.73 [−8.06, +6.23]	.920
Baseline arm	273 responses	261/273 (95.60%)	.973	—	—
Reminder arm	273 responses	258/273 (94.51%)	.962	—	—
Rater A	273 pairs	—	—	−.73 (point)	—
Rater B	273 pairs	—	—	−1.47 (point)	—
Qwen/BM25	90 pairs	—	—	−1.11 [−14.44, +12.22]	1.000
Llama/BM25	93 pairs	—	—	0.00 [−10.75, +10.75]	1.000
Qwen/MiniLM-L6	90 pairs	—	—	−1.11 [−14.44, +12.22]	1.000

Table 2: **Rater- and setting-level robustness.** Panel A shows evaluator-minus-human arm error; its Reminder–Baseline difference is CD. Panel B reports pre-adjudication agreement over 546 responses, then rater-specific or adjudicated score- ≥ 2 effects (Reminder minus Baseline). Rater rows are point estimates; setting CIs resample the same 18 case clusters (seed 2703).

Outcome source	Δ_j pp [95% CI]	CD _j pp [95% CI]
Human reference	−1.67 [−8.3, +5.0]	0.00 (Ref.)
Qwen2.5-3B	−40.00 [−48.3, −31.7]	−38.33 [−47.5, −29.2]
Qwen2.5-7B	−23.33 [−31.7, −15.0]	−21.67 [−30.0, −13.3]
Qwen2.5-14B	−15.00 [−21.7, −8.3]	−13.33 [−20.8, −5.8]
DeepSeek-v4-Flash	−3.33 [−8.3, +1.7]	−1.67 [−6.7, +3.3]
Gemini 2.5 Flash	−3.33 [−8.3, +1.7]	−1.67 [−5.8, +2.5]

Table 3: **Pooled OOD same-ID replication** on 60 pairs across four domains. Effects use score- ≥ 2 labels; Δ_j is Reminder minus Baseline and $CD_j = \Delta_j - \Delta_h$. CIs use 10,000 domain-stratified pair-bootstrap resamples (seed 2703); the supplement gives counts and Holm tests.

either use an evaluator that fails the item screen or lack a matched human contrast; none supports the headline intervention claim. Pooled acknowledgement likewise motivates an arm-specific audit but does not validate a lexical rule for conditioning the comparison.

6 Implications for Valid Inference

Practical use of the correction. Estimate the machine contrast on the larger machine-only set and correct it with evaluator–human residuals from a probability-sampled set of labeled pairs. Choose γ on held-out data or by cross-fitting, with $\gamma = 0$ as the human-only fallback.

GAVEL makes two local decisions: whether an evaluator passes the item-level screen and whether its familywise CD confidence region lies inside $[-\epsilon, \epsilon]$. Passing both supports use for labels and for the audited contrast. Passing only the item screen supports item labeling; passing only the contrast screen supports only the audited comparison. Passing neither limits raw machine effects to sensitivity analysis. Every decision is local to the frozen rubric, response population,

Outcome	Generator	Dense	Δ [95% CI]	p
Human T3	Qwen	L6	−.111 [−.556, +.370]	1.000
	Qwen	L12	+ .000 [−.333, +.333]	1.000
	Llama	L6	−.111 [−.481, +.278]	1.000
	Llama	L12	−.370 [−.648, −.074]	.215
	Mistral	L6	−.056 [−.500, +.370]	1.000
	Mistral	L12	−.074 [−.500, +.352]	1.000
Exposure [†]	Qwen	L6	+ .323 [+ .261, +.373]	.031
	Qwen	L12	+ .215 [+ .175, +.261]	.031

Table 4: **Dense–BM25 contrasts under two outcomes.** Human T3 rows are ordinal differences ($N = 54/\text{cell}$); CIs resample 18 cases and p is Holm-adjusted over six contrasts, so the Llama/L12 pointwise CI is not familywise significant. [†]Exposure rows are Qwen cache-poison proportions ($N = 54/\text{retriever}$; BM25 .403) with descriptive, unadjusted six-domain sign-flip p -values.

and contrast family.

All evaluators with an item-level audit fail that screen. Qwen also fails the contrast screen because its distortion intervals are far from zero. DeepSeek and the stack have smaller observed CD, but intervals crossing zero do not establish containment inside a tolerance. Gemini has only an OOD contrast audit, so we assign it no item-screen status. Conversely, an evaluator that passes an item screen would still require a same-ID CD audit before supporting a treatment claim. The two decisions are logically and empirically distinct.

Prediction-powered correction uses machine pair scores only as a control variate; probability-sampled human pairs estimate their residual error. A held-out or cross-fitted rule may choose $\gamma = 0$, recovering human-only inference when

machine scores do not reduce variance. Proposition 2 therefore guarantees validity, not empirical efficiency, which we do not test.

Designing the Human Sample

The scientific tolerance comes before the label budget. Investigators first specify the largest distortion ϵ that would leave the substantive conclusion unchanged. A pilot set of complete human-labeled pairs then estimates the cluster-level variability of $d^h - \gamma d^j$. That residual variability, the target interval width, and the number of independent case clusters determine the required sample. When several generators, retrievers, or domains contribute to the claim, the probability sample should be stratified over those factors and combined with frozen design weights.

Sampling complete pairs is essential. Labeling whichever arm appears uncertain changes the target population and can bias both the human effect and the correction. If adaptive acquisition is necessary, each pair needs a known inclusion probability and both arms must enter together. Evaluator selection, γ tuning, and final inference must use disjoint splits or cross-fitting. These rules let a weak evaluator contribute variance reduction when its pair differences are informative without allowing it to define the estimand.

The CD audit and Equation 5 answer different questions. The first asks whether a raw machine contrast can substitute for the human contrast; the second asks whether machine scores can reduce the label cost of estimating it. A source may fail certification yet remain useful as a control variate, while a small observed CD does not guarantee variance reduction.

A Contrast Certificate

A report that uses an automated evaluator for a comparison should expose enough information to reconstruct the claim, not merely the score. A contrast certificate records: (i) the response population, pair key, conditions, and direction of subtraction; (ii) the rubric, threshold, evaluator version, arm counts, and common denominator; and (iii) $\hat{\Delta}_h$, $\hat{\Delta}_j$, and $\widehat{CD}_j$, together with intervals, multiplicity, tolerance, sampling design, and any prediction-powered weight. A pooled confusion matrix can accompany this record but cannot replace it.

This format also prevents scope drift. Certification for a Reminder–Baseline contrast says nothing about a dense–BM25 contrast, a new prompt template, or a different response population. Any of those changes creates a new measurement claim and requires a new same-ID audit. GAVEL therefore produces local certificates rather than a permanent ranking of evaluators.

7 Reproducibility

Stable record IDs connect cases, conditions, responses, and evaluator outputs. Hashes and strict joins detect duplicate or missing arms, response drift, and split leakage. We release replayable analysis records for the 194-response reference, the 273-pair six-channel audit, the 60-pair OOD replication, and the 486-response three-family retrieval benchmark, together with code for CD analysis, rater sensitivity, prediction-

powered correction, and equivalence testing. The supplement states which response text and first-pass forms remain restricted.

8 Limitations

Empirical quadrants. None of the automated sources with an item-level audit passes that screen, so the study does not observe the “reliable labels, distorted contrast” quadrant. Gemini has only an OOD contrast audit. Proposition 1 establishes that item calibration cannot identify CD; it does not establish how often every quadrant occurs in practice. Small observed CD for DeepSeek, the stack, or Gemini is not equivalence without a chosen tolerance and adequate power.

Human measurement and PPI assumptions. Adjudicated labels are a reference measurement, not error-free latent truth. Our interval conditions on those labels; separate-rater point estimates are a sensitivity check, not an interval for a broader annotator population. Prediction-powered validity requires a probability sample of complete human-labeled pairs and a fixed or cross-fitted tuning rule. Selective annotation without known inclusion probabilities can bias the correction.

Scope. The main study tests one reminder intervention on controlled multi-turn RAG cases; the OOD replication contains 60 pairs across four operational domains. Neither estimates deployment prevalence. Secondary prompt and retrieval analyses are underpowered for small effects and are not evidence of invariance. Contrast distortion is an observed judge–condition interaction; it does not identify attention, intent, lexical triggering, or another internal mechanism.

Outcome and design dependence. The CD identity applies to any fixed numerical outcome, but Proposition 1’s sharp envelope uses binary labels and the score-at-least-2 threshold. Ordinal or continuous outcomes require their own scale and tolerance. The paired audit estimates an average difference over generated response pairs, not a causal effect on one already-realized response. Finally, the controlled cases deliberately stress contradictory evidence. They expose measurement failure but do not determine how often it occurs in open-world traffic.

9 Conclusion

Pooled item calibration does not identify contrast validity. On identical pairs, humans estimate a small Reminder effect of $-.73$ points, with an interval too wide to establish equivalence, while three Qwen evaluators report reductions of 14.65–38.10 points. Their ranking by distortion magnitude reappears in a pooled four-domain replication. GAVEL addresses this mismatch by auditing contrast distortion and deriving a probability-sampled correction from human-labeled pairs. The practical rule is direct: calibrate for labels, and audit for comparisons.

Ethical Statement

We release no live data; public artifacts support auditing but may also inform retrieval attacks.

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